

Koushica Bosadi Ulaganathan

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Professional Summary

Software Engineer with 4+ years of experience building and shipping production web applications across the full stack. Proven track record in React, Vue, Node.js, and TypeScript, from architecting component systems and leading code review to delivering backend APIs and data-driven dashboards. Experienced in distributed team leadership, legacy-to-modern codebase migrations, and integrating AI tooling into engineering workflows.

Skills & Qualifications

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3

Front-End: React.js, Vue.js (2 & 3), Next.js, Redux, Zustand, D3.js, Tailwind CSS, CSS Modules, Framer Motion

Back-End & Data: Node.js, Express.js, MongoDB, Mongoose, MySQL, RESTful APIs, JWT

AI & Tooling: GitHub Copilot, Claude Code, Gemini API, HuggingFace Inference API, Git, GitHub Actions, CI/CD, Jest, Webpack, Babel, React Testing Library, Docker, Figma, JIRA

CMS & Other: Drupal, Omni CMS, Agile/Scrum

Professional Experience

One Community Global | Software Engineer | *Remote*

Oct 2024 – Jun 2025

- Built an interactive geospatial map feature using React and Leaflet.js to visualize employee locations, track contributions, and support distributed team management across global project areas.
- Developed a data reporting dashboard using D3.js, generating pie charts and bar charts to visualize individual employee performance reports and overall team contribution metrics.
- Served as lead code reviewer for a distributed team of 15+ engineers, enforcing TypeScript standards and architectural guidelines, contributing to a 35% reduction in production bugs.
- Integrated GitHub Copilot into the team workflow for unit test generation and boilerplate reduction, improving development velocity by approximately 25%.
- Engineered a dynamic SEO metadata system and real-time payment and wishlist features, improving platform discoverability and user engagement.
- Implemented a full dark mode architecture using Tailwind CSS, ensuring accessibility compliance and visual consistency across the application.
- Adopted CSS Modules for feature-level style encapsulation, eliminating class name conflicts and keeping component styles independently maintainable across the codebase.

University of North Texas, DESRC | Graduate Assistant | *Denton, TX*

Sep 2023 – May 2024

- Maintained and updated departmental websites using Drupal and Omni CMS across three student resource programs serving hundreds of students per semester.
- Designed and implemented a student ID tracking system for the campus Food Pantry, enabling data-driven restocking decisions and usage reporting.
- Managed day-to-day operations of three student programs, Food Pantry, Suit-Up Closet, and Mean Green Grad Gown, including volunteer training and scheduling.
- Produced digital content and managed social media accounts, organizing campus events focused on food insecurity awareness and available student resources.

Vezham Technologies | Software Engineer | *Remote*

Jun 2022 – Jan 2023

- Built the frontend of a CMS product from the ground up using React.js, TypeScript, and Tailwind CSS, contributing across UI design, feature development, and code review.
- Introduced TypeScript strict mode across the frontend codebase, reducing runtime errors by 35% and improving long-term code reliability.
- Developed a custom in-browser markdown editor in React, which became a core component of the content creation workflow.
- Established Jest test coverage and strengthened CI/CD pipelines, improving release stability and deployment confidence.
- Contributed to backend performance improvements using Node.js and TypeScript, reducing page load times by 30%.

- Revamped frontend architecture by migrating from legacy JavaScript to Vue.js, cutting page load times by 40% and eliminating key performance bottlenecks across the product.
- Diagnosed and resolved critical memory leak issues causing page freezes on high-volume list views, reducing load times by 42% and stabilizing application performance.
- Developed a real-time chat feature using Vue.js and Vuex, enabling modular state management and improving feature delivery speed through a scalable architecture.
- Built a visitor tracking system handling 2,000+ users, monitoring user behavior patterns to optimize engagement and prioritize resource allocation.
- Designed reusable UI component libraries, including forms, calendars, and dropdowns, standardizing development and ensuring visual consistency across product areas.
- Collaborated with product managers, designers, and backend engineers in an agile environment, contributing to sprint planning, peer code reviews, and cross-functional feature delivery.

Projects

PantryPal: *React 18, TypeScript, Node.js, Express, MongoDB, Tailwind CSS, Zustand, JWT, Spoonacular API*

Full-stack recipe discovery platform that matches meals to available pantry ingredients in real time. Features ingredient-based matching with percentage scoring, drag-and-drop weekly meal planner, auto-generated shopping lists, user authentication, dark/light mode, and infinite scroll, backed by 365,000+ recipes.

Error Decoder: *React 18, TypeScript, Node.js, Express, Tailwind CSS, Google Gemini API*

AI-powered developer tool that transforms cryptic error messages and stack traces into plain-English explanations with root cause analysis and fix suggestions. Supports 20+ programming languages and frameworks, with structured JSON output via prompt engineering and graceful fallback handling.

Shica Talks: *React, Vite, JavaScript, HuggingFace Inference API*

AI-powered chat application with real-time responses, full markdown rendering, XSS-safe sanitization, and keyboard shortcuts, built on HuggingFace's open-source Llama model.

Education

M.S. Information Systems & Technologies, GPA 4.0 / 4.0

May 2024

University of North Texas - *GreatGrad, Spring 2024*

B.E. Electronics & Communication Engineering, GPA 3.2 / 4.0

Apr 2019

Anna University, India